

test 4

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1. The functions f and fg are defined by $f(x) = e^{(x^2 + 3)}$ for $x < 0$ and $fg(x) = e^{(2x)}$ for $x > 3/2$. Explain why f^{-1} exists. [1 mark]

2. Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1} . [5 marks]

3. Hence find and simplify an expression for $g(x)$. [2 marks]

4. The functions f and g are defined, for $x > 0$, by $f(x) = (2x^2 - 1)/(3x)$ and $g(x) = 1/x$. Find and simplify an expression for $fg(x)$. [2 marks]

5. On the axes, sketch the graph of $y = |(x^2 - 4)(x - 2)|$, stating the intercepts with the coordinate axes. [4 marks]

6. Find the possible values of the constant k for which $|(x^2 - 4)(x - 2)| = k$ has exactly 4 different solutions. [2 marks]

7. Find the values of x for which y exists. [2 marks]

8. A function h is defined by $h(x) = 2x/(3x+1)$ for the largest possible domain. State the domain of h . [1 mark]

9. Explain how the diagram shows the relationship between f and f^{-1} . [1 mark]

10. Using your answer to part (ii) state the relationship between the functions f and f^{-1} . [1 mark]

11. Find $f(f(x))$, giving your answer in its simplest form. [2 marks]
